

PYTHON PROGRAMMING

COURSE OBJECTIVES:

- To acquire programming skills in core Python
- To learn Strings and function
- To develop object oriented skills in Python
- To comprehend various Python Packages
- To develop web applications using Django

UNIT I

Introduction : Fundamental ideas of Computer Science - Strings, Assignment and Comments - Numeric Data types and Character sets - Expressions - Loops and Selection Statements: Definite iteration: the for Loop - selection: if and if-else statements - Conditional iteration: the while Loop

UNIT II

Strings and Text Files: Accessing Characters and substrings in strings - Data encryption-Strings and Number systems- String methods - Text - Lists and Dictionaries: Lists - Dictionaries - Design with Functions: A Quick review - Problem Solving with top-Down Design - Design with recursive Functions - Managing a Program's namespace - Higher-Order Functions

UNIT III

Design with Classes: Getting inside Objects and Classes - Data-Modeling Examples - Building a New Data Structure - The Two - Dimensional Grid - Structuring Classes with Inheritance and Polymorphism-GraphicalUserInterfaces-The Behavior of terminal-Based programs and GUI-Based programs - Coding Simple GUI-Based programs - Windows and Window Components - Command Buttons and responding to events

UNIT IV

Working with Python Packages: NumPy Library-Ndarray- Basic Operations - Indexing, Slicing and Iteration - Array manipulation - Pandas - The Series - The DataFrame - The Index Objects - Data Vizualization with Matplotlib- The Matplotlib Architecture -Pyplot- The Plotting Window - Adding Elements to the Chart - Line Charts - Bar Charts - Pie charts

UNIT V

Django: Installing Django- Building an Application - Project Creation - Designing the Data Schema - Creating an administration site for models - Working with QuerySets and Managers - Retrieving Objects - Building List and Detail Views